

Initial Thermal Model Friction Imbalance Model – Summary

Model Idea:

A model to represent the effect of thermal differences on the ice and the motion of a curling stone.

The pressure of the stone on the ice melts it, different sections of ice have different temperatures and therefore different values of the coefficient of friction. In this model, I have split the edges of the stone up into 4 temperature bands, each temperature band with a different coefficient of friction assigned to each temperature band.

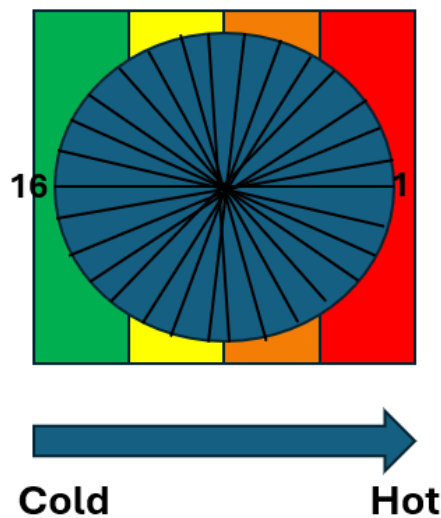


Figure 1: A diagram showing 4 temperature bands on the ice.

Then, I am splitting the edge of the stone into n points where n is an even integer, as shown in figure 1.

The coefficient of friction is determined for a point and the point directly opposite it. The difference between these two coefficients is calculated using the equation $\Delta\mu = \mu_1 - \mu_2$ where $\Delta\mu$ is the change in coefficient of friction values between the two points and is equal to the coefficient of friction of the first point minus the second. This is repeated so that the difference between all pairs of opposite points is calculated.

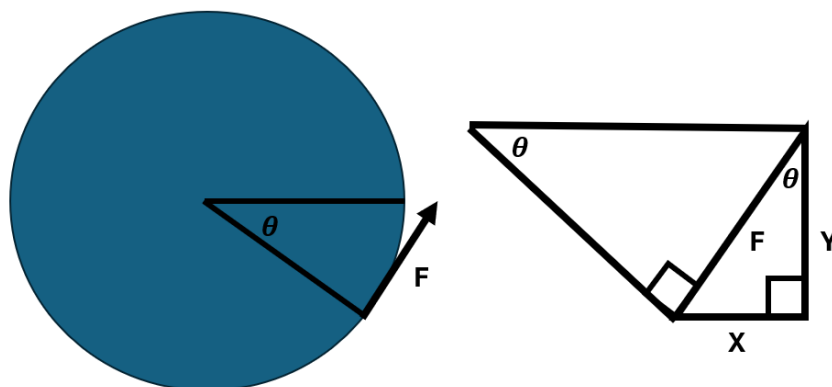


Figure 2: A diagram showing how to determine the direction of the overall force.

The force is calculated for each change in coefficient of friction using the equation $\Delta F = \Delta\mu N$ where ΔF is the change in force and N is the normal force. The components in the X and Y direction of this force for each pair of points are determined. The equations to determine the X and Y components of the force can be calculated using figure 2, $F_x = F \sin(\theta)$ and $F_y = F \cos(\theta)$. Where F is the overall force for each pair of points and θ is the angle shown in figure 2.

Then, the x and y force for each pair of points is summed together to get the overall force in both the x and y directions on the stone caused by the temperature bands.

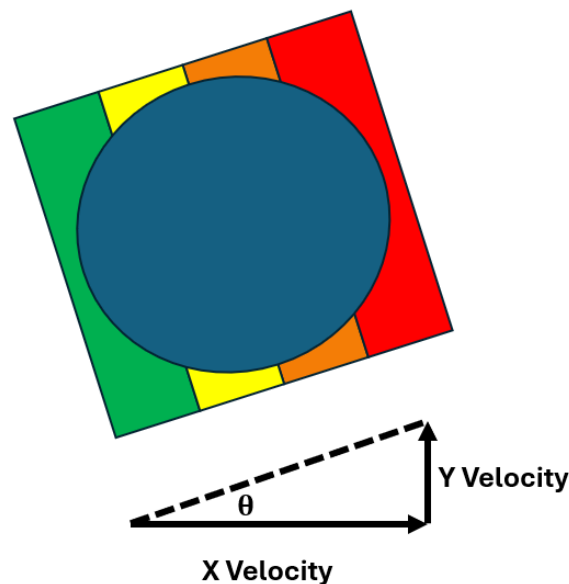


Figure 3: A figure showing how the temperature band rotates with the translational velocity direction

As the stone moves, it gains y velocity therefore, its overall translational velocity is no longer in the x direction. I am modelling it so that the temperature bands rotate to point in the direction of the translational velocity, as shown in figure 3.

Assumptions:

- The ice sheet is smooth, it has no scratches or pebbles
- There are 4 discrete temperature bands
- Coefficient of friction has a linear relationship to ice temperature
- Temperature bands rotate with the translational velocity direction

Predictions:

This model predicts a curl in the direction opposite to the angular rotation of the stone. E.g. a clockwise spin of the curling stone leads to a curl in the positive Y direction. It also predicts that there will be a very small curl. A real curling stone curls in the same direction as its angular rotation, which is opposite to the direction to the curl produced by the thermal model.

Results:

This model was then made in python with the results below.

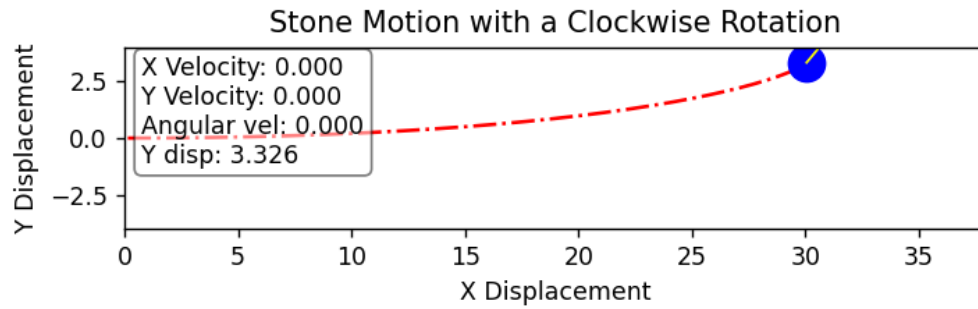


Figure 4: A graph showing the predicted motion of a curling stone when it has a clockwise rotation.

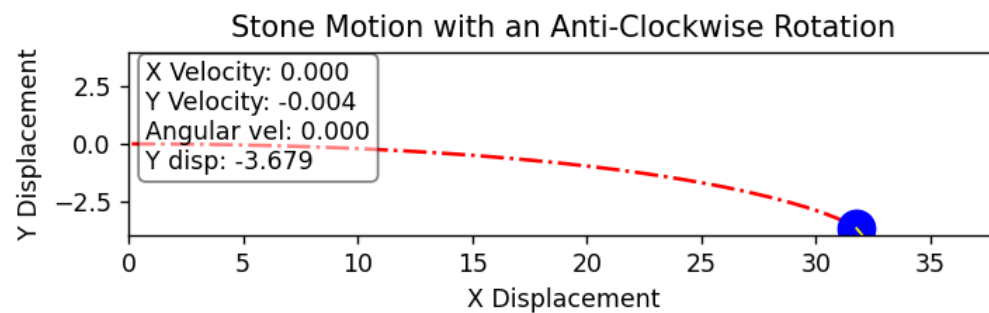


Figure 5: A graph showing the predicted motion of a curling stone when it has an anticlockwise rotation.

Conclusion:

Overall, this model predicts that a friction imbalance caused by thermal differences in the ice can cause a curling stone to curl on the ice, in the direction opposite to the angular rotation which is the opposite direction to what we see in real curling stones. For a clockwise rotation, it produces a Y displacement of 3.326 metres and a Y displacement of -3.679 metres for an anticlockwise rotation. Where this model fails is the magnitude of the Y displacement as its magnitude should be smaller, as well as the two paths should be symmetric.

Future improvements:

- 1) Using coefficient of friction and temperature values that are not guessed to achieve a more realistic curl.
- 2) A function of temperature around the edge of the stone instead of 4 discrete temperature bands.
- 3) Taking the presence of pebbles and scratches into account.